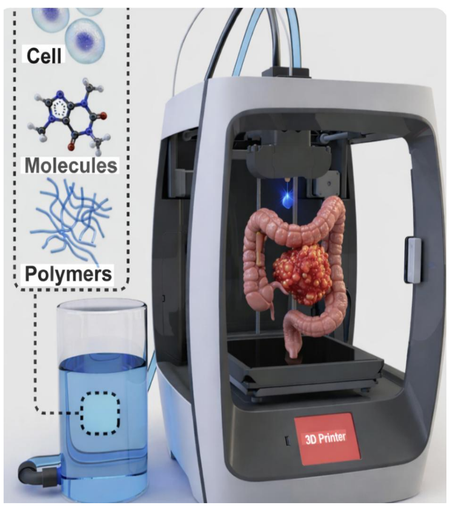
GHOST TUMOR: a new platform to study Colorectal Tumor Microenvironment

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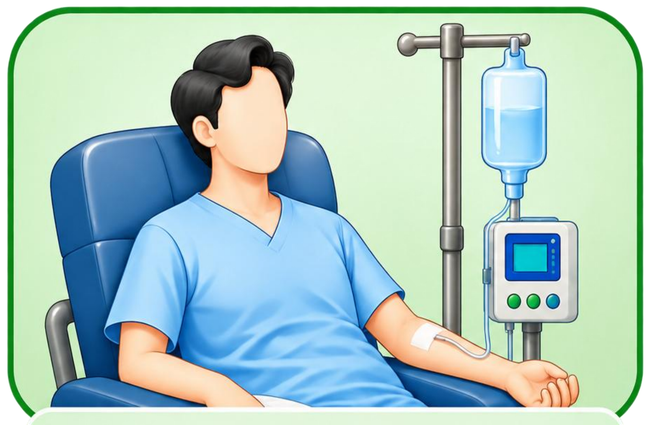
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Project

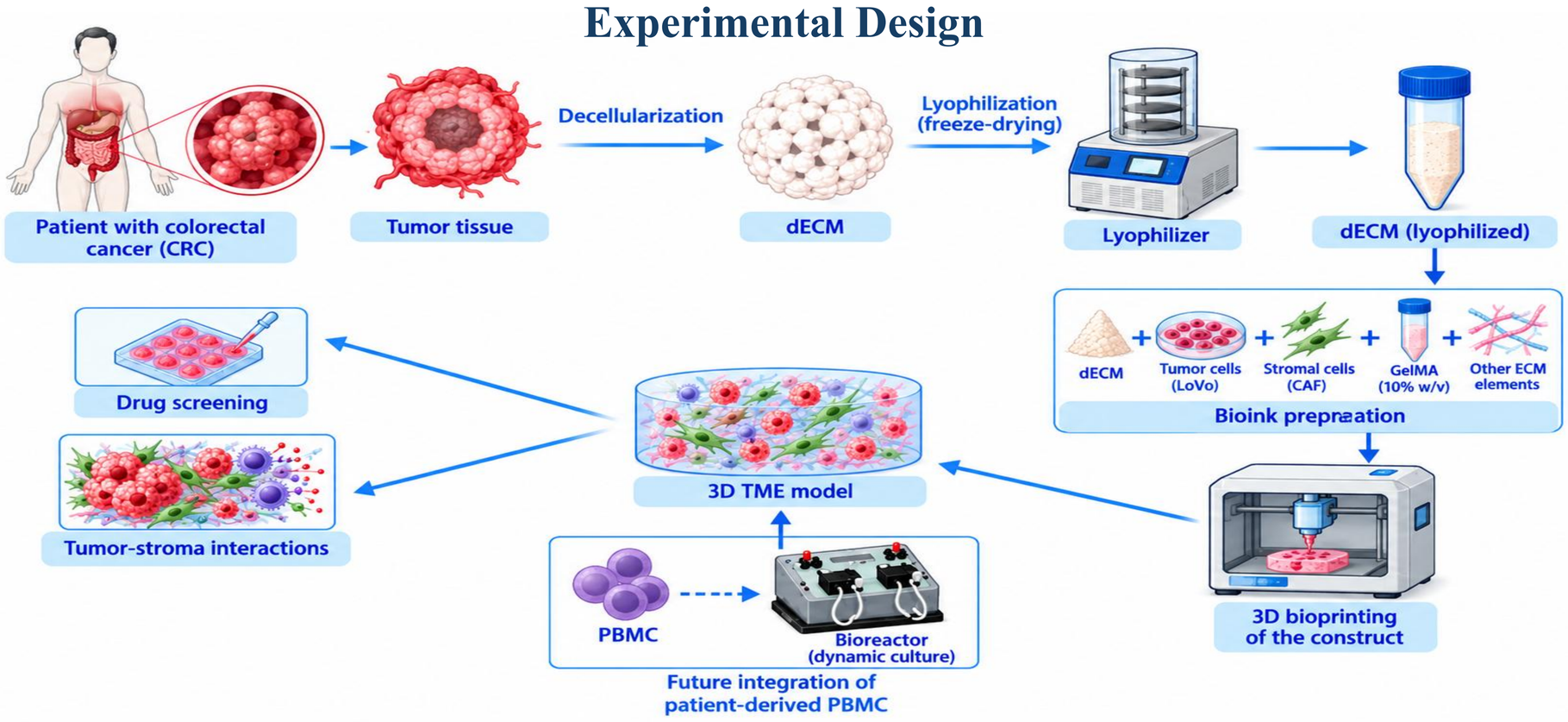


Every tumor is embedded in a scaffold of proteins - the extracellular matrix (ECM) - that governs its growth, invasion and, above all, the immune system's ability to attack it. Tumor cells do more than grow: they **rebuild their own home**, making the matrix denser and stiffer, like a fortress that shields the tumor from the immune system. **Ghost Tumor** brings this discovery to the exhibition through two mutually reinforcing elements: the **"ghost organ" -** an actual decellularized tissue scaffold, the translucent ECM that remains after all cells have been removed - displayed in a direct comparison of **normal matrix vs. tumor matrix**; and an **interactive station** where visitors build the tumor microenvironment with their own hands. They choose the matrix components, adjust its stiffness, seed tumor cells, fibroblasts and immune cells, administer a drug or immunotherapy, and observe in real time whether the tumor grows, whether T cells can infiltrate it or remain "excluded," and whether the treatment works.

The experience translates a real laboratory process - **decellularization → recellularization → therapeutic testing -** into a hands-on activity that can be understood in two minutes, making visible how a tumor model is built and why it matters for personalized medicine.

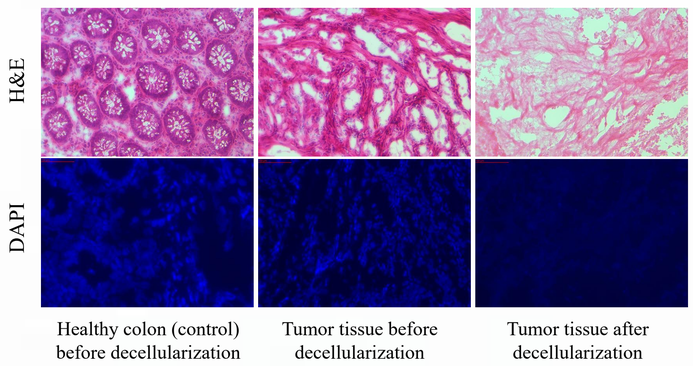


Experimental Design

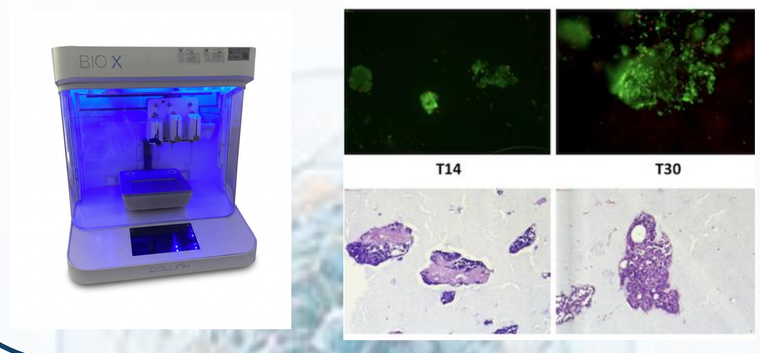


Results

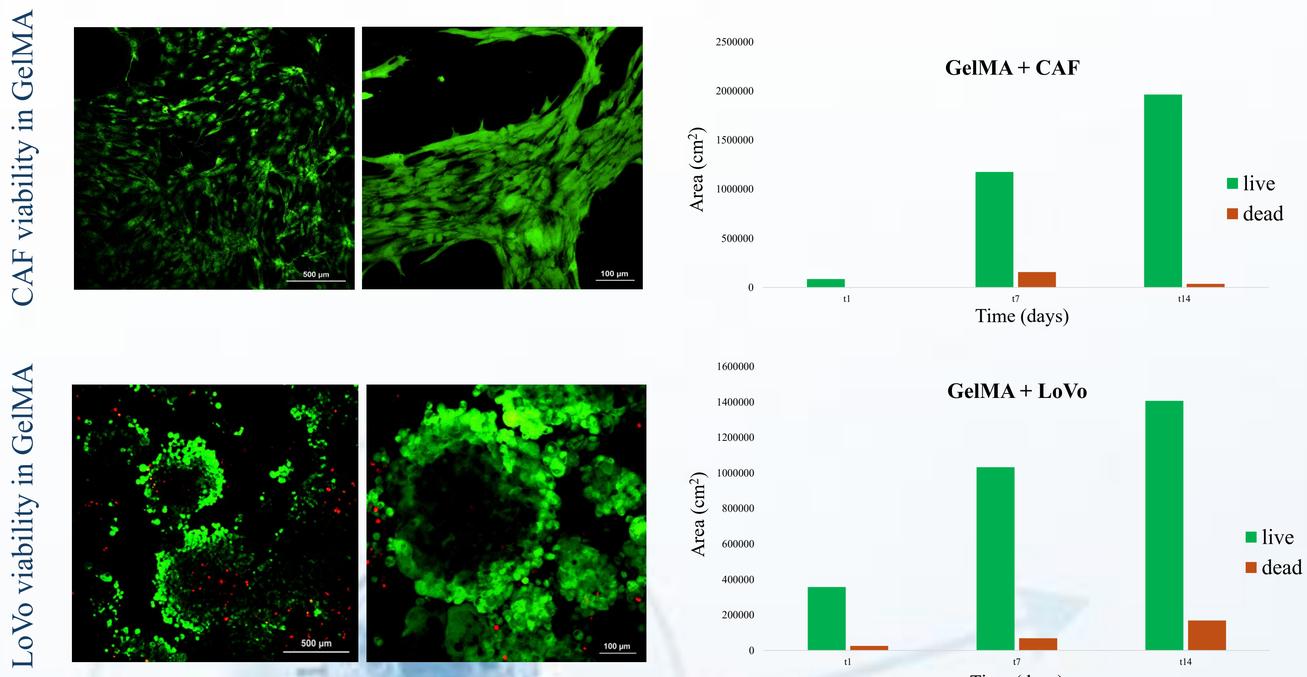
Characterization of Patient-Derived dECM



3D printed tumor



Biocompatibility GelMA - Cells



Representative Live/Dead staining of LoVo and CAF cells cultured in GelMA.

Conclusions

The "ghost tumor" model aims to draw the attention of researchers and the public to the importance of identifying new platforms for studying how the immune system or new drugs can influence tumor growth without using animal models.